

# HAT-P-3b

R-1390

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-3\_160207 · created 2026-08-01T02:53:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457425.8662 +/- 0.0021 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.083 +/- 0.02                  |
| Transit depth (Rp/Rs)^2                           | 0.68 +/- 0.33 [%]               |
| Orbital Inclination (inc)                         | 86.73 +/- 1.49                  |
| Airmass coefficient 1 (a1)                        | 0.9819 +/- 0.0097               |
| Airmass coefficient 2 (a2)                        | 0.0121 +/- 0.0029               |
| Scatter in the residuals of the lightcurve fit is | 0.99 %                          |
| Best Comparison Star                              | #2 - [235, 368]                 |
| Optimal Aperture                                  | 3.69                            |
| Optimal Annulus                                   | 5.1                             |
| Transit Duration (day)                            | 0.074 +/- 0.022                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.083 ± 0.02 vs 0.1063 (deviation 1.14σ) |
| Duration fit vs published | 0.074 d vs 0.0867 d (deviation 0.56σ)    |
| Mid-time O–C              | -5.2 min                                 |
| Depth SNR                 | 4.1                                      |
| Residual scatter          | 0.99 %                                   |

## Light curve

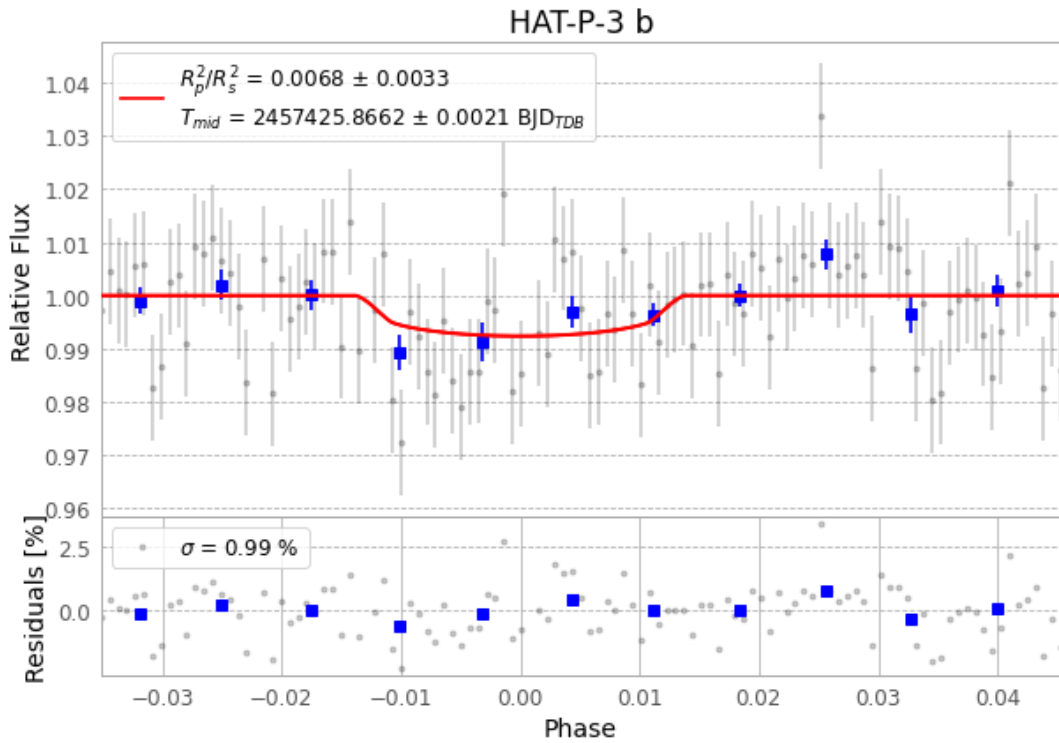


Figure 1 — FinalLightCurve\_HAT-P-3 b\_2016-02-06.png (SHA-256 a628798e74c2c7d6...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 145], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 f3d3d9e81bdebc17...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

113 FITS frames (74 MB), HATP-3160207061512.FITS → HATP-3160207115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-160207.log (1bc5415f46cb...), FinalLightCurve\_HAT-P-3 b\_2016-02-06.png (a628798e74c2...), FinalLightCurve\_HAT-P-3 b\_2016-02-06.csv (643bc8a88e6f...)

## Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 02:53Z.